

DataStax Developer Day
DSE Operations / Security

Configuring DSE

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Making configuration files FUN!!!



- Let’s look at important configuration files
- Why they matter

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Important configuration files



cassandra.yaml



cassandra-rackdc.properties

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cassandra.yaml

- Cassandra nodes read this file on start-up
  - restart nodes for changes to take effect

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cassandra.yaml

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- Located in the following directories
  - Cassandra package installations: /etc/dse/cassandra
  - Cassandra tarball installations: <INSTALL_LOCATION>/resources/cassandra/conf

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- cluster_name (Default: “Test Cluster”)
  - Once cluster_name is set it cannot be changed unless the cluster is rebuilt
- listen_address (Default: localhost)
- native_transport_address (Default: localhost)
- seeds (Default: “127.0.01”)
  - You must have at least one seed node per datacenter

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cassandra-rackdc.properties

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cassandra-rackdc.properties

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- dc (Default: “dc1”)
  - Once dc is set it cannot be changed unless the DC is rebuilt
- rack (Default: “rack1”)
  - Once rack is set it cannot be changed unless the nodes are moved to a new rack

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Honorable mentions

- DSE
  - dse.yaml
  - If you are trying to use something not common to both cassandra and DSE then the dse.yaml is the first place to look.
  - Settings
    - Security
      - Auth
      - Encryption
    - DSE Search
    - Performance
    - DSE Analytics
    - Advanced replication

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  - Settings
    - Security
      - Auth
      - Encryption
    - DSE Search
    - Performance
    - DSE Analytics
    - Advanced replication
- Cassandra
  - cassandra-env.sh
  - jvm.options
- OpsCenter
  - <cluster_name>*.conf
  - Agent
    - address.yaml
- Studio
  - configuration.yaml

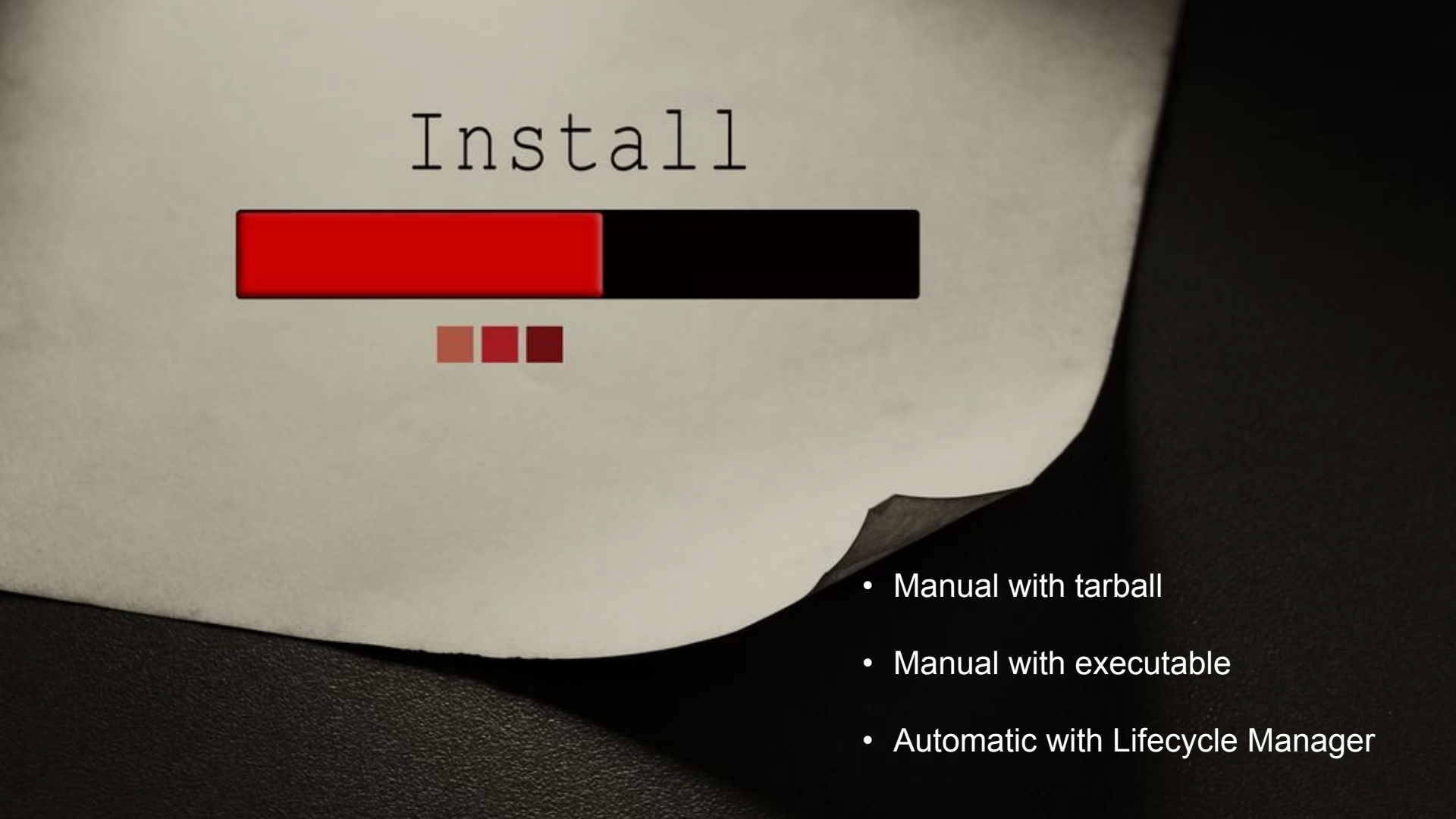
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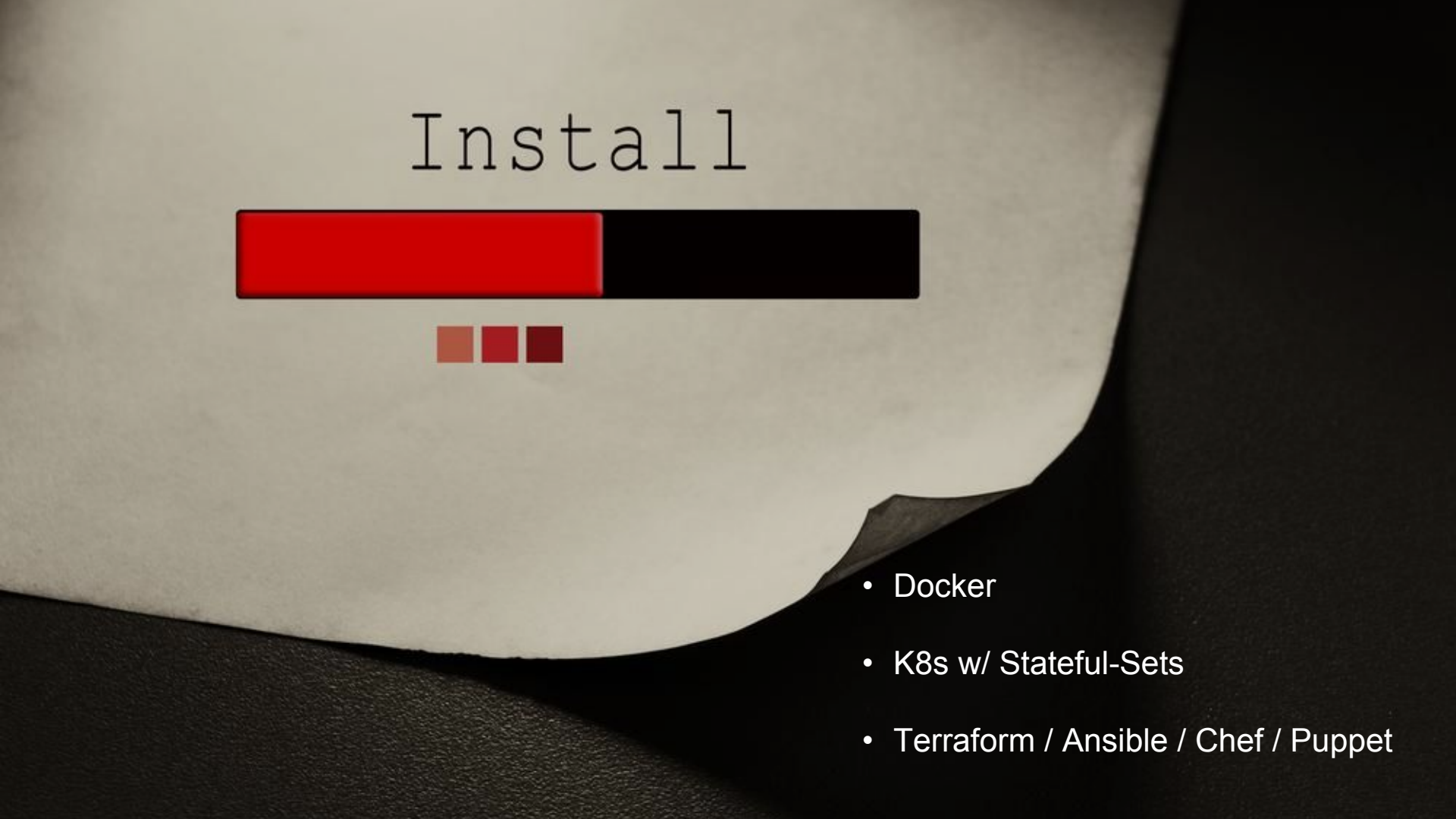
Installing DSE

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- Manual with tarball
- Manual with executable
- Automatic with Lifecycle Manager



- Docker
- K8s w/ Stateful-Sets
- Terraform / Ansible / Chef / Puppet

Exercise

Open up DataStax Studio and the notebook:
DSE Operations & Security: Installing DSE



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- Install DSE and add a node to an existing DSE cluster
- Use LCM to install the DSE cluster from scratch

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Monitoring and Performance

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Coming from a relational database?

- How you tune your database for performance can be very different from what you may be used to
  - Even scary, like really scary

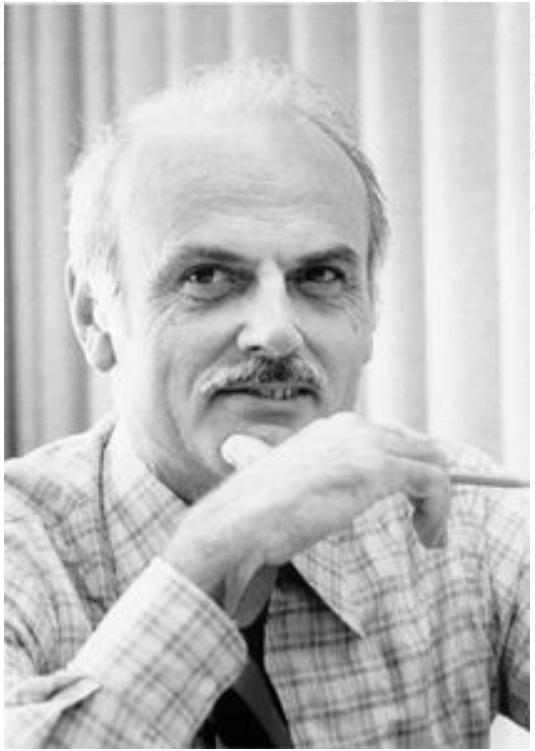


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DSE Performance

- Storing data "correctly"
- Performance optimizations come afterwards
- We tune our queries using explain plans



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DSE Performance

What do we do with DSE?

- Designed for OLTP operations
- Queries are meant to be fast!
  - No joins
  - Data is distributed across different nodes
  - Replicas mean that DSE has multiple nodes to read from
- Don't need explain plans

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DSE Performance

How this gets complicated…

This is a distributed system!

- Data… data everywhere
- Workloads and resources on different instances

Read path and write path… know this and how read / write performance is affected by this

- Compaction
- Anti-entropy repair and node sync operations

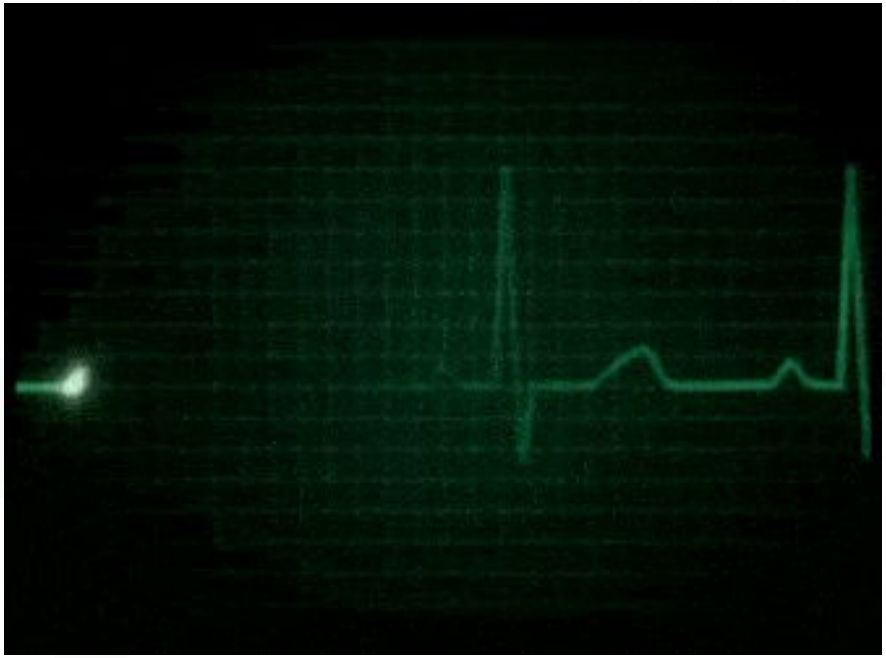
Data modeling

- The irony - **you're optimizing the data model now, not your queries**

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Cassandra Monitoring



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```
Keyspace : killrvideo
        Read Count: 13516
        Read Latency: 0.09195841965078425 ms
        Write Count: 0
        Write Latency: NaN ms
        Pending Flushes: 0
                Table: comments_by_user
                SSTable count: 1
                Space used (live): 1443744
                Space used (total): 1443744
                Space used by snapshots (total): 0
                Off heap memory used (total): 5960
                SSTable Compression Ratio: 0.5464805748827252
                Number of partitions (estimate): 4510
                Memtable cell count: 0
                Memtable data size: 0
                Memtable off heap memory used: 0
                Memtable switch count: 3
                Local read count: 0
                Local read latency: NaN ms
                Local write count: 0
                Local write latency: NaN ms
                Pending flushes: 0
                Percent repaired: 0.0
                Bytes repaired: 0.000KiB
                Bytes unrepaired: 2.423MiB
                Bytes pending repair: 0.000KiB
                Bloom filter false positives: 0
                Bloom filter false ratio: 0.00000
                Bloom filter space used: 5656
                Bloom filter off heap memory used: 5648
                Index summary off heap memory used: 0
                Compression metadata off heap memory used: 312
                Compacted partition minimum bytes: 73
                Compacted partition maximum bytes: 3973
                Compacted partition mean bytes: 617
                Average live cells per slice (last five minutes): NaN
                Maximum live cells per slice (last five minutes): 0
                Average tombstones per slice (last five minutes): NaN
                Maximum tombstones per slice (last five minutes): 0
                Dropped Mutations: 0
                Failed Replication Count: null
```

Cassandra Monitoring

The Open Source way

- Low level stuff
  - JMX Mbeans - lots of metrics
  - Nodetool commands
    - Might be running scripts to capture, collect, and aggregate output
    - Cron jobs

DSE Performance Service

Queries and statistics collected and saved in tables

- DSE will monitor your queries
- Slow queries? Will let you know what's going on
- Just pull out of the database and use wherever you want



DSE Performance Service

Queries and statistics collected and saved in CQL tables

- What else…
  - CQL Slow Query Log
  - Latency metrics on tables
  - Per node and cluster lifetime metrics by table and keyspace
  - Recent and lifetime statistics about tables, sstable count, partition size, etc
  - Read / write activity by per-client and per-node
  - DSE Search bottlenecks
  - DSE Analytics resource usage
  - Monitor specific DSE Analytics applications

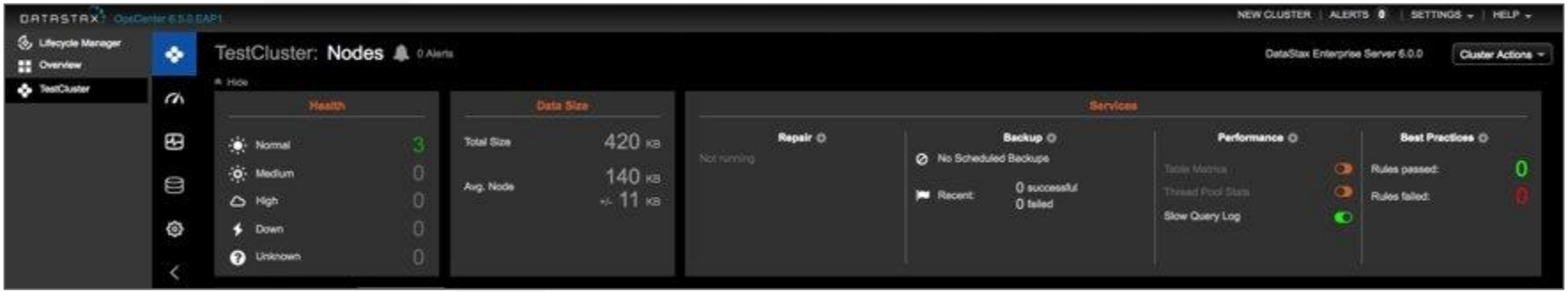
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OpsCenter overview



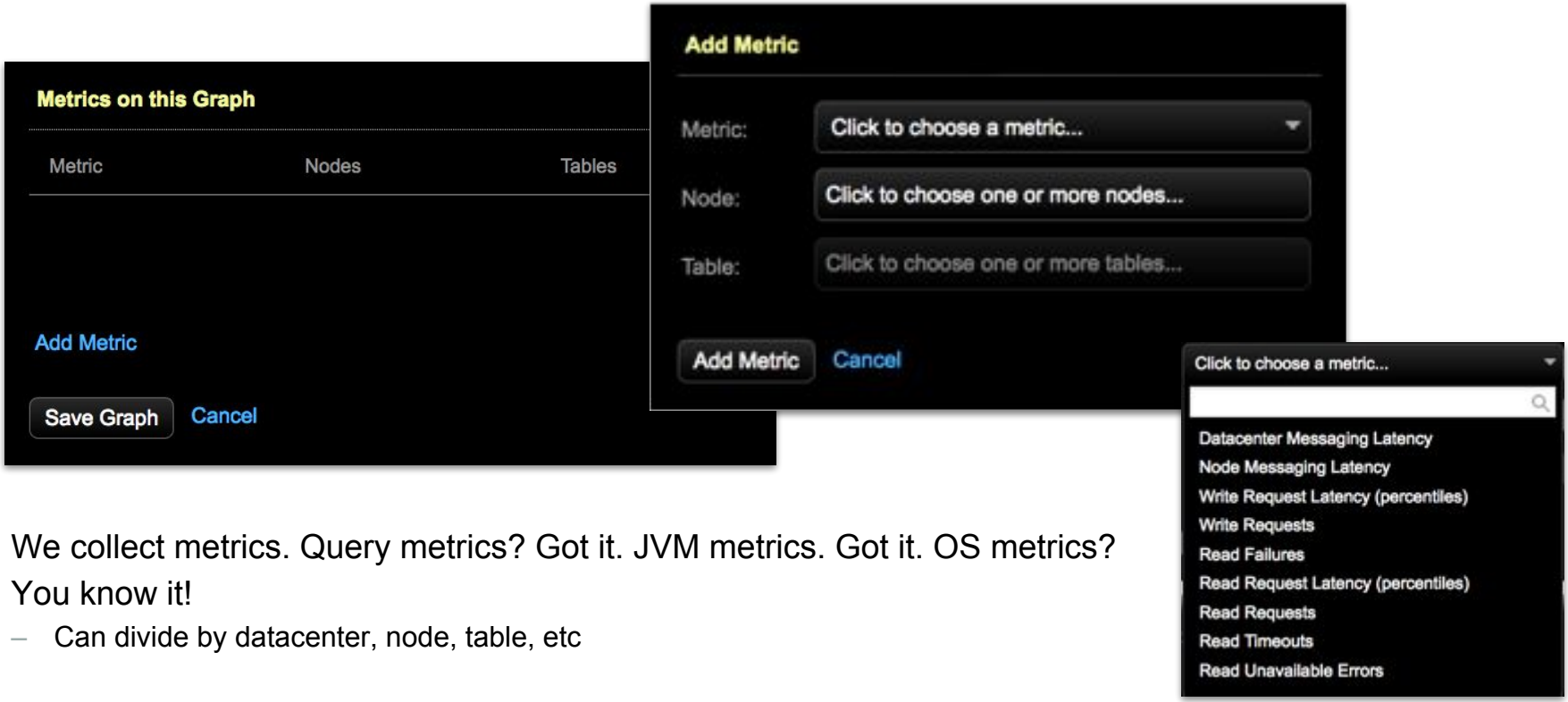
- Browser-based DSE cluster tool for:
  - Configuring
  - Monitoring
  - Managing
- Two major components tied together:
  - OpsCenter Monitoring - monitoring and management
  - Life Cycle Manager (LCM) - mostly configuration and deployment



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DSE Performance



- We collect metrics. Query metrics? Got it. JVM metrics. Got it. OS metrics? You know it!
  - Can divide by datacenter, node, table, etc

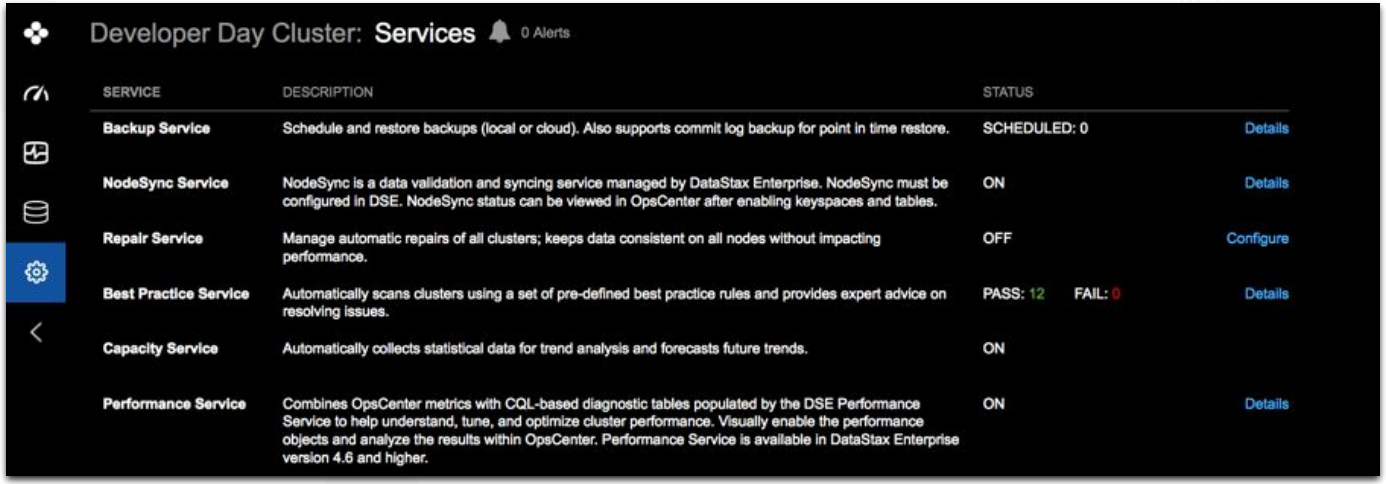
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DSE Performance

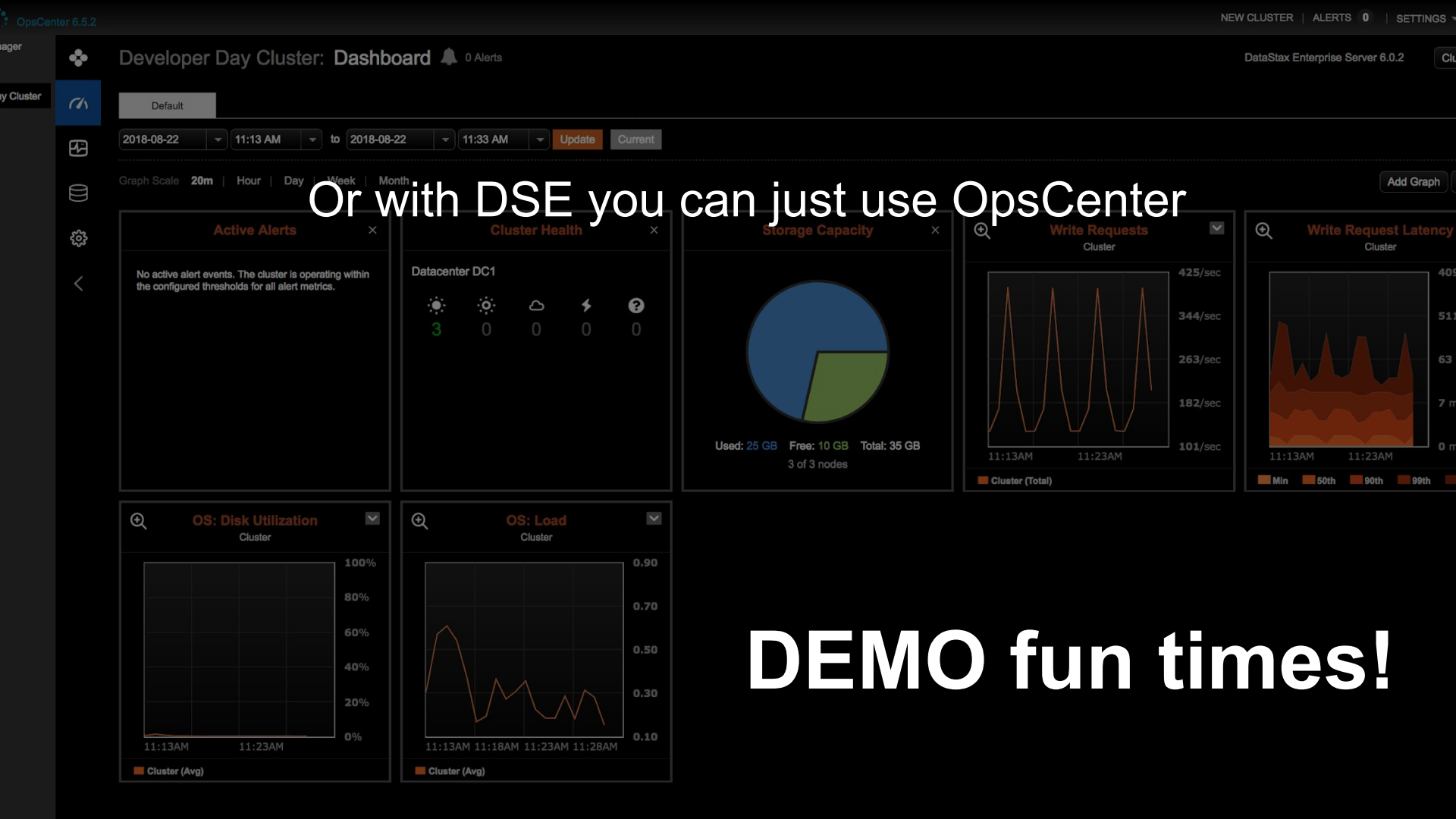
OpsCenter Services

- Want to know when your nodes are slowing down? We'll send you a message
- Best practices service, performance service, lots of stuff
- Even capacity planning



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OpsCenter 6.5.2
NEW CLUSTER | ALERTS 0 | SETTINGS
Developer Day Cluster: Dashboard
0 Alerts
DataStax Enterprise Server 6.0.2
Default
2018-08-22
11:13 AM
to
2018-08-22
11:33 AM
Update
Current
Graph Scale 20m | Hour | Day | Week | Month
Add Graph
Or with DSE you can just use OpsCenter
Active Alerts
No active alert events. The cluster is operating within the configured thresholds for all alert metrics.
Cluster Health
Datacenter DC1
3 0 0 0 0
Storage Capacity
Used: 25 GB Free: 10 GB Total: 35 GB
3 of 3 nodes
Write Requests
Cluster
425/sec
344/sec
263/sec
182/sec
101/sec
11:13AM
11:23AM
Cluster (Total)
Write Request Latency
Cluster
11:13AM
11:23AM
Min
50th
90th
99th
OS: Disk Utilization
Cluster
100%
80%
60%
40%
20%
0%
11:13AM
11:23AM
Cluster (Avg)
OS: Load
Cluster
0.90
0.70
0.50
0.30
0.10
11:13AM 11:18AM 11:23AM 11:28AM
Cluster (Avg)
DEMO fun times!

Security

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Security overview

Bad security results in data breaches. DSE has a number of features that can be used to secure your data at all stages.

- End to end encryption
- Data auditing
- LDAP integration
- Kerberos integration
- Role based access control
- Row-level access control

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Encryption

Encryption keeps your data safe so that even if people get their hands on it they will not be able to use it.



Auditing

Auditing lets you keep track of what is going on with your system and who has been accessing it.



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Auditing

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AUTH - Logs login events

DML - Logs insert, update, delete and other data manipulation language (DML) events

DDL - Logs object and role create, alter, drop, and other data definition language (DDL) events

DCL - Logs grant, revoke, create role, drop role, and list roles events

QUERY - Logs all queries

https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secAuditTOC.html

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Authentication and Authorization

What do these terms mean?

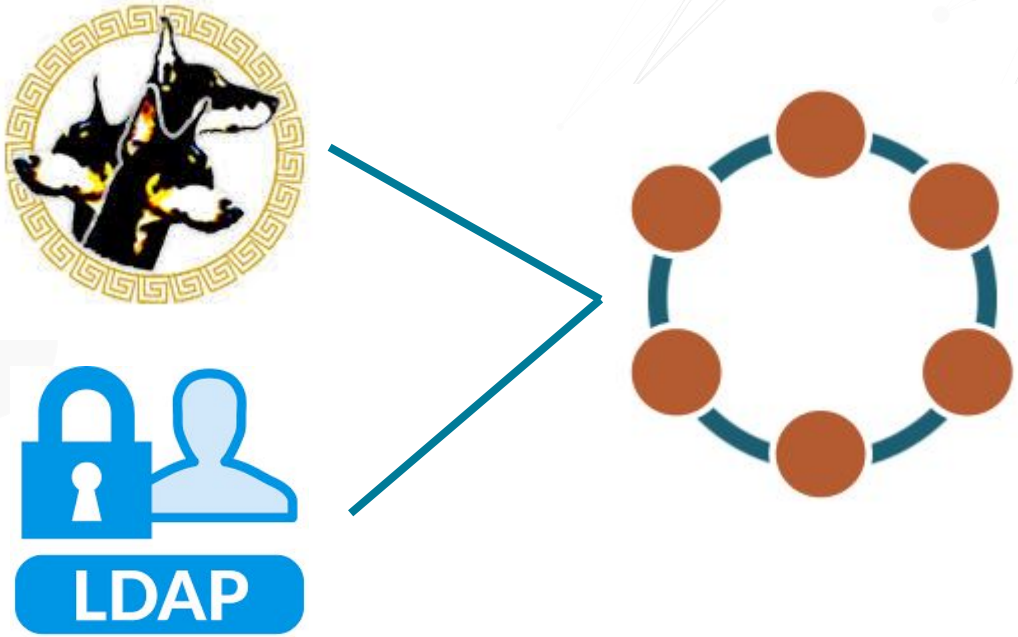
- Authentication controls who can access the cluster
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secDSEUnifiedAuthAbout.html
- Authorization controls what they can do on the cluster
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secRbacAbout.html

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LDAP & Kerberos

Account management is made easier when integrated with the existing systems you use.



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Exercise

Open up DataStax Studio and the notebook:
DSE Operations & Security: Securing DSE



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- Use the Lifecycle Manager to enable authentication and authorization
- Also use the Lifecycle Manager to enable audit logging
- Finally, use Lifecycle Manager to turn on in-flight encryption

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Useful resources

- Security checklist and features
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secChecklists.html#secChecklists
- Encryption
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/encryptClientNodeSSL.html
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secEncryptEnable.html
- Auditing
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secAuditTOC.html
- Authentication
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secAuthTOC.html
- Role based access control
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secRbacAbout.html
- Row-level access control
  - https://docs.datastax.com/en/dse/6.0/dse-admin/datastax_enterprise/security/secRlac.html

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Some resources to learn more

- DataStax Academy
  - DS210 - DataStax Enterprise Operations with Apache Cassandra
    - https://academy.datastax.com/resources/ds210-datastax-enterprise-operations-apache-cassandra
- DataStax Academy Slack
  - #courses
  - #opscenter

https://academy.datastax.com



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Thanks!



Adron Hall, *Developer Advocate*

Programming languages & Areas of Interest: Java/ C#, Javascript, Go. Solving complex distributed systems problems.
Interesting Projects: Rocket & missile telemetry, science research workflow.
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Twitter Handle: @Adron
Twitch Handle: @AdronHall



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Interesting Projects:
Github Handle: @mando222
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